

Loops in Python

Loops are used to **repeat a block of code multiple times** until a condition is met.

The main types are:

1. **For loop** – sequential traversal (when number of iterations is known).
 2. **While loop** – repeats until a condition becomes false.
 3. **Nested loops** – loops inside other loops.
 4. **Loop control statements** – break, continue, pass.
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1. For Loop

- Used to iterate over sequences like range, lists, tuples, strings, dictionaries.
- Best when we **know how many times** to run the loop.

Syntax

```
for variable in range(start, end, step):
```

```
    # code block
```

Example 1 – Using range()

```
n = 4
```

```
for i in range(0, n):
```

```
    print(i)
```

Output:

```
0
```

```
1
```

```
2
```

```
3
```

Example 2 – Traversing a list

```
fruits = ["apple", "banana", "cherry"]
```

```
for fruit in fruits:
```

```
    print(fruit)
```

Output:

```
apple
```

```
banana
```

```
cherry
```

2. While Loop

- Used when we **don't know in advance** how many times the loop should run.
- The loop continues **as long as condition is True**.

Syntax

while condition:

 # code block

Example

```
i = 0
while i < 5:
    print(i)
    i += 1
```

Output:

```
0
1
2
3
4
```

3. Loop Control Statements

These statements change the normal flow of loops.

Statement	Purpose
break	Exits the loop completely
continue	Skips the current iteration and moves to the next
pass	Does nothing, acts as a placeholder

Example – break

```
for i in range(10):
    if i == 5:
        break
    print(i)
```

Output:

0
1
2
3
4

Example – continue

```
for i in range(5):
```

```
    if i == 2:
```

```
        continue
```

```
    print(i)
```

Output:

0
1
3
4

Example – pass

```
for i in range(5):
```

```
    if i == 2:
```

```
        pass # Does nothing
```

```
    print(i)
```

Output:

0
1
2
3
4

4. Nested Loops

- Loops inside other loops.
- Commonly used in **patterns, tables, matrices**.

Example – Pattern

```
n = 5
for i in range(1, n+1):
    for j in range(i):
        print("*", end="")
    print()
```

Output:

```
*
**
***
****
*****
```

5. Practice Questions (Unsolved)

- a) Find sum of digits of a number (e.g., $a=123 \rightarrow 1+2+3=6$)
 - b) Reverse a string (e.g., "abcd" $\rightarrow$ "dcba")
 - c) Check whether a string is palindrome (e.g., "abcba" $\rightarrow$ same when reversed)
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6. Pattern Printing Examples

Right-Angled Triangle

```
n = 5
for i in range(1, n+1):
    print('*' * i)
```

Output:

```
*
**
***
****
*****
```

Inverted Triangle

```
n = 5
```

```
for i in range(n, 0, -1):
```

```
    print('*' * i)
```

Output:

```
*****
```

```
****
```

```
***
```

```
**
```

```
*
```

Number Pattern

```
rows = 5
```

```
for i in range(1, rows+1):
```

```
    for j in range(i):
```

```
        print(i, end="")
```

```
    print("")
```

Output:

```
1
```

```
22
```

```
333
```

```
4444
```

```
55555
```

Reverse Number Pattern

```
rows = 5
```

```
num = rows
```

```
for i in range(rows, 0, -1):
```

```
    for j in range(i):
```

```
        print(num, end="")
```

```
    print("")
```

Output:

```
55555
```

5555

555

55

5